

CS & IT ENGINEERING



Computer Network

IPv4 Address

Lecture No. – 03

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Recap of Previous Lecture



Topic

Network Size

Topic

Network Mask



Topics to be Covered



Topic

Classful IPv4 Address

Topic

Subnetting *



ABOUT ME



Hello, I'm **Abhishek**

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Example 6 :-

[MSQ]



#Q. Consider a class-less IPv4 network, the broadcast address (special IPv4 address) of this network is "175 . 175 . 175 . 255", which is/are can be a network address (special IPv4 address) of this network ?

~~A~~

175 . 160 . 0 . 0

~~B~~

175 . 175 . 0 . 0

☒ C

175 . 175 . 160 . 0

☒ D

175 . 175 . 175 . 0

Ans: C & D



Network Directed Broadcast Address = 175.175.175.255

10101111 10101111 10101111 11111111

Min^m 20 bit NetID

Max^m 12 bit hostID

Min^m 2 bit HostID

Network Address =

10101111 10101111 10101111 11111100
 30 bit NetID 2 bit HostID

10101111 10101111 10100000 00000000
 20 bit NetID 12 bit HostID

✗ 175.160.0.0 = 175.160.00000000.00000000

✗ 175.175.0.0 = 175.175.00000000.00000000

✓ 175.175.160.0 = 175.175.10100000.00000000
12 bit HostID

✓ 175.175.175.0 = 175.175.10101111.00000000
24 bit NetID 8 bit HostID

[MCQ]

[GATE-2010][2 Mark]



#Q. Suppose computers A and B have IP addresses "10 . 105 . 1 . 113" and "10 . 105 . 1 . 91" respectively and they both use the same netmask N. Which of the values of N given below should not be used if A and B should belong to the same network?

- ☒ A 255 . 255 . 255 . 0
- ☒ B 255 . 255 . 255 . 128
- ☒ C 255 . 255 . 255 . 192
- ☐ D 255 . 255 . 255 . 224

113 → 01110001

91 → 01011011

192 → 11000000

224 → 11100000

Ans: D

10.105. 1.113
255.255. 255.224

10.105. 1. 91
255.255. 255.224

10.105. 1. 96

← Not Same →

10.105. 1. 64

10.105. 1.113
255.255. 255.192

10.105. 1. 91
255.255. 255.192

10.105. 1. 64

← Same →

10.105. 1. 64

10.105. 1.113
255.255.255.128

10.105. 1. 91
255.255.255.128

10.105. 1. 0 ← *same* → 10.105. 1. 0

10.105. 1.113
255.255.255. 0

10.105. 1. 91
255.255.255. 0

10.105. 1. 0 ← *same* → 10.105. 1. 0

10.105.1.113 = 10.105.1.01110001

10.105.1.91 = 10.105.1.01011011

255.255.255.0 = 255.255.255.00000000

255.255.255.128 = 255.255.255.10000000

255.255.255.192 = 255.255.255.11000000

255.255.255.224 = 255.255.255.11100000

[MCQ]

[GATE-2003][2 Mark]



#Q. The subnet mask for a particular network is "255 . 255 . 31 . 0". Which of the following pairs of IP addresses could belong to this network?

- ☒ A 172 . 57 . 88 . 62 and 172 . 56 . 87 . 233
- ☒ B 10 . 35 . 28 . 2 and 10 . 35 . 29 . 4
- ☒ C 191 . 203 . 31 . 87 and 191 . 234 . 31 . 88
- ☒ D 128 . 8 . 129 . 43 and 128 . 8 . 161 . 55

Ans: D

255.255.31.0

|||||·|||||·000||||·00000000

X

```

172. 57. 88. 62
255.255. 31.  0
-----
172. 57 .    . 0
  
```

```

172. 56 . 87. 233
255.255. 31.  0
-----
172. 56 .    . 0
  
```

X

```

10. 35. 28.  2
255.255. 31.  0
-----
10. 35. 28 . 0
  
```

```

10. 35. 29.  4
255.255. 31.  0
-----
10. 35. 29 . 0
  
```

← Not Same →

X

```

191.203.31.87
255.255.31.0
-----
191.203.   .0
  
```

```

191.234.31.88
255.255.31.0
-----
191.234.   .0
  
```

✓

```

128.  8.129.43
255.255.31.0
-----
128.  8. 1 .0
  
```

```

128.  8.161.55
255.255.31.0
-----
128.  8. 1 .0
  
```

← Same →



Topic : IPv4 Address



Problem : How many bits are assigned for network id field ?

→ IPv4 address are categorized into two type :

1. Classful IPv4 Address

→ Static Assignment : NetID field size defined implicitly



2. Classless IPv4 Address

→ Dynamic Assignment : Need to define NetID field size explicitly

*CIDR



Topic : Classful IPv4 Address

→ Classful IPv4 Address are of five types :

1. Class A } → NetID = 8 bit
2. Class B } → NetID = 16 bit
3. Class C } → NetID = 24 bit
4. Class D } Special IP
5. Class E } Address

classful
or
class A
or
class B
or
class C

↓
[default
class-less]



Topic : Class A IPv4 Address

→ IP Address = 0.....
[Start with Zero bit]

→ IP address range :

0 . 0 . 0 . 0

to

127 . 255 . 255 . 255

[0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0]

[0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]

2^{31} Address



Topic : Class A IPv4 Address



Network ID : 8 bits

Host ID : 24 bits

Number of Networks = 2^7 networks → 128
or
126

Number of Hosts per Network = $(2^{24} - 2)$ hosts

→ Default Network Mask for Class A IPv4 network : 255.0.0.0



Topic : Class B IPv4 Address

→ IP Address = 10.....
[Start with One & Zero bit]

→ IP address range :

128.0.0.0

to

191.255.255.255

[10000000 00000000 00000000 00000000]

[10111111 11111111 11111111 11111111]

2^{30} IP Address



Topic : Class B IPv4 Address



Network ID : 16 bits

Host ID : 16 bits

Number of Networks

$$= 2^{14} \text{ networks}$$

Number of Hosts per Network

$$= (2^{16} - 2) \text{ hosts}$$

→ Default Network Mask for Class B IPv4 network : 255.255.0.0



Topic : Class C IPv4 Address



→ IP Address = 110.....
[Start with One, One & Zero bit]

→ IP address range :

192.0.0.0

to

223.255.255.255

[11000000 00000000 00000000 00000000]

[11011111 11111111 11111111 11111111]

2²⁹ IP Address



Topic : Class C IPv4 Address



Network ID : 24 bits

Host ID : 8 bits

Number of Networks

$$= 2^{24} \text{ networks}$$

Number of Hosts per Network

$$= (2^8 - 2) \text{ hosts}$$

→ Default Network Mask for Class C IPv4 network : 255.255.255.0

[MCQ]

[GATE-2012][1 Mark]



#Q. In the IPv4 addressing format, the number of networks allowed under Class C addresses is :

A 2^7

B 2^{14}

✓ C 2^{21}

~~D 2^{24}~~

Ans: C



Topic : Class D IPv4 Address

→ IP Address = 1110.....
[Start with One, One, One & Zero bit]

→ IP address range :

224.0.0.0

to

239.255.255.255

[11100000 00000000 00000000 00000000]

[11101111 11111111 11111111 11111111]

228 IP Add.



Topic : Class D IPv4 Address



- Used for “Multicasting”
- Class D IPv4 address can not be a host IP address
- It is used as “Group Id” by IGMP $\Rightarrow$ N/w layer Protocol
[Internet Group Message Protocol]





Topic : Class E IPv4 Address



→ IP Address = 1111.....
[Start with four One bit]

→ IP address range :

240 . 0 . 0 . 0

to

255 . 255 . 255 . 255

[11110000 00000000 00000000 00000000]

[11111111 11111111 11111111 11111111]

→ Reserved for future use

228 IPAdd.



Topic : Classful IPv4 Address

Class	IP	NetID Bits	First Octet	Efficiency
<u>Class A</u>	<u>0</u>	<u>8</u> - bits	0 to 127	<u>50 %</u>
<u>Class B</u>	<u>10</u>	<u>16</u> - bits	128 to 191	<u>25 %</u>
<u>Class C</u>	<u>110</u>	<u>24</u> - bits	192 to 223	<u>12.5 %</u>
<u>Class D</u>	<u>1110</u>	N. A	224 to 239	6.25 %
<u>Class E</u>	<u>1111</u>	N. A	240 to 255	6.25 %

100 %



Topic : Classless IPv4 Address

→ Number of bits for network id field are assigned 'Dynamically'

→ No any prefix bits for IPv4 address
[Unlike Classful IPv4 Address] $\Rightarrow [2^{32} \text{ IP Add.}]$

→ Classless IPv4 Address : $P.Q.R.S/x$

→ Prefix or Slash notation [x] represent "number of network id bits"



Topic : Subnetting



→ Dividing (logically) a network into smaller manageable sub-networks

→ Sub-network (subnet) : Clustering of hosts inside a network

→ Clustering of hosts based on some bits of host identifier (HostId) field
[In practice, most significant bits of host identifier]



Topic : Subnetting



Suppose,

Net ID field size is **27 bits**

Host ID field size is **5 bits**

No. of IP Add. = 2^5

Max^m no. of hosts
= $(2^5 - 2)$ hosts

Host ID (5 bits)

0	0	0	0	0
0	0	0	0	1
0	0	0	1	0
0	0	0	1	1
0	0	1	0	0
0	0	1	0	1
0	0	1	1	0
0	0	1	1	1
0	1	0	0	0
0	1	0	0	1
0	1	0	1	0
0	1	0	1	1
0	1	1	0	0
0	1	1	0	1
0	1	1	1	0
0	1	1	1	1

1	0	0	0	0
1	0	0	0	1
1	0	0	1	0
1	0	0	1	1
1	0	1	0	0
1	0	1	0	1
1	0	1	1	0
1	0	1	1	1
1	1	0	0	0
1	1	0	0	1
1	1	0	1	0
1	1	0	1	1
1	1	1	0	0
1	1	1	0	1
1	1	1	1	0
1	1	1	1	1



Topic : Subnetting



Suppose,

Net ID field size is 27 bits

Host ID field size is 5 bits

Host ID (5 bits)

0	0	0	0	0
0	0	0	0	1
0	0	0	1	0
0	0	0	1	1
0	0	1	0	0
0	0	1	0	1
0	0	1	1	0
0	0	1	1	1
0	1	0	0	0
0	1	0	0	1
0	1	0	1	0
0	1	0	1	1
0	1	1	0	0
0	1	1	0	1
0	1	1	1	0
0	1	1	1	1

1	0	0	0	0
1	0	0	0	1
1	0	0	1	0
1	0	0	1	1
1	0	1	0	0
1	0	1	0	1
1	0	1	1	0
1	0	1	1	1
1	1	0	0	0
1	1	0	0	1
1	1	0	1	0
1	1	0	1	1
1	1	1	0	0
1	1	1	0	1
1	1	1	1	0
1	1	1	1	1



Topic : Subnetting

Suppose,

Net ID field size is 27 bits

Host ID field size is 5 bits

Host ID (5 bits)

0 0 0 0 0

0 0 0 0 1

0 0 0 1 0

0 0 0 1 1

0 0 1 0 0

0 0 1 0 1

0 0 1 1 0

0 0 1 1 1

0 1 0 0 0

0 1 0 0 1

0 1 0 1 0

0 1 0 1 1

0 1 1 0 0

0 1 1 0 1

0 1 1 1 0

0 1 1 1 1

1 0 0 0 0

1 0 0 0 1

1 0 0 1 0

1 0 0 1 1

1 0 1 0 0

1 0 1 0 1

1 0 1 1 0

1 0 1 1 1

1 1 0 0 0

1 1 0 0 1

1 1 0 1 0

1 1 0 1 1

1 1 1 0 0

1 1 1 0 1

1 1 1 1 0

1 1 1 1 1





Topic : Subnetting



→ Before subnetting, IP address having two sections :

1. Network Identifier (Net ID) : x bits

2. Host Identifier (Host ID) : y bits ←

→ Size of IP address field = (x + y) bits



Topic : Subnetting



→ After subnetting, IP address having three sections :

1. Network Identifier (Net ID) : x bits
2. Sub-network Identifier (Subnet ID) : y_1 bits
3. Host Identifier (Host ID) : y_2 bits

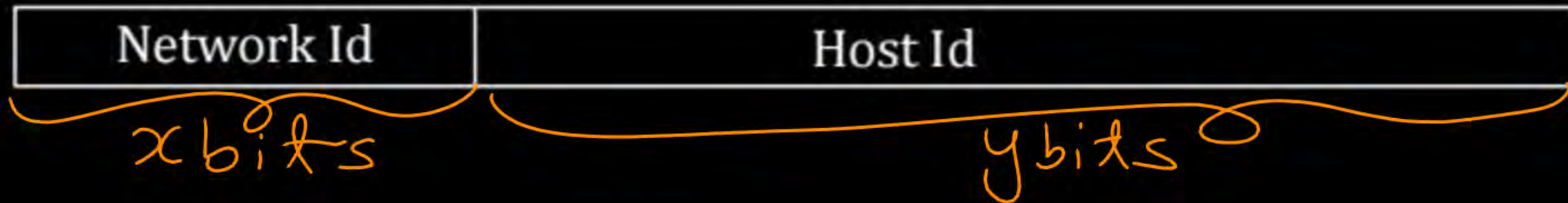
→ Size of IP address field = $(x + y_1 + y_2)$ bits = $(x + y)$ bits
[$y = y_1 + y_2$]



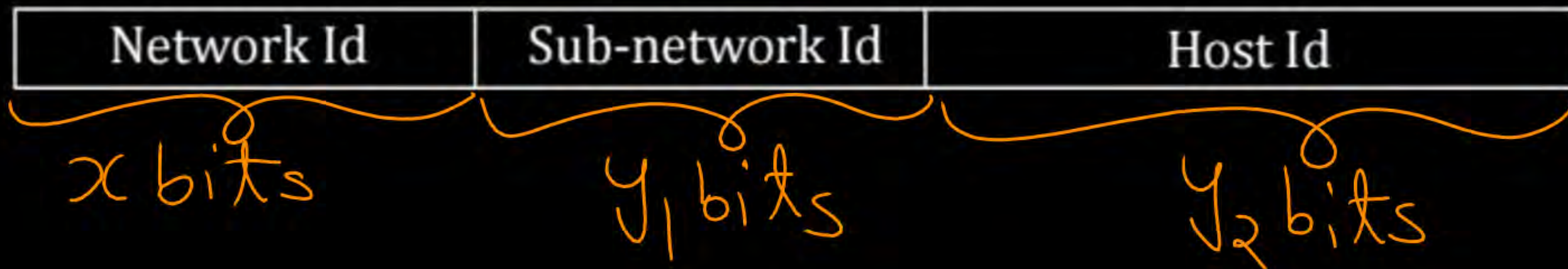
Topic : Subnetting



Before Subnetting



After Subnetting





Topic : Sub-network Address

$\Rightarrow$ Subnet Address



→ Special IP address (32 bits)

→ Used to represent a sub-network

NetID field = [As Assigned]

Subnet ID = [Anything] $\Rightarrow$ [All zero ... All ones]

HostID field = All Zero Bits

Net Id	Subnet Id	Host Id [0 0 0 0 0 0 0 0]
--------	-----------	-------------------------------------

x bits

y_1 bits

y_2 bits



Topic : Sub-network Broadcast Address

- Special IP address (32 bits)
- Used to broadcast a packet to all hosts belongs to a sub-network

NetID field = [As Assigned]

Subnet ID = Anything $\Rightarrow$ [All zero to All ones]

HostID field = All One Bits

Net Id	Subnet Id	Host Id [1 1 1 1 1 1 1 1]
--------	-----------	-----------------------------------

x bits

y_1 bits

y_2 bits



Topic : Host IP Address



→ Host IP address (32 bits)

→ Used to identify a host uniquely world wide

NetID field (x bits) = [As Assigned]

Subnet ID field (y_1 bits) = Anything $\Rightarrow$ [All zero to All one]

HostID field (y_2 bits) = Any thing
[Except all zero and all one bits]



Example 13 :- [H.W.]

Class C network (Network Address : 200 . 200 . 200 . 0), with 3-bit subnetting.

Network Address : 200 . 200 . 200 .

Broadcast Address : 200 . 200 . 200 .
(Network Directed)

Network Mask :

Before Subnetting :-

First Host IP Address : 200 . 200 . 200 .

Last Host IP Address : 200 . 200 . 200 .

Network Size =



2 mins Summary



Topic

Classful IPv4 Address ✓

Topic

Subnetting ⇒



THANK - YOU

